TTK4150 - Assignment 1

Problem 1

a)

$$x_1 = \begin{bmatrix} 6 \\ -10 \end{bmatrix} \quad x_2 = \begin{bmatrix} -5 \\ -4 \\ 3 \end{bmatrix}$$

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{\frac{1}{p}}, \quad p \in [1, \infty]$$

$$\|x_1\|_1 = |6| + |10| = 16$$

$$\|x_2\|_1 = |-5| + |-4| + |3| = 12$$

$$\|x_1\|_2 = \sqrt{|6|^2 + |-10|^2} = \sqrt{136} = 2\sqrt{34}$$

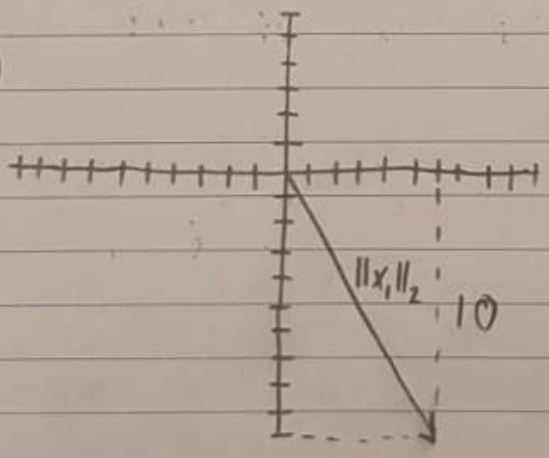
$$\|x_2\|_2 = \sqrt{|-5|^2 + |-4|^2 + |3|^2} = \sqrt{50} = 5\sqrt{2}$$

$$\|x\|_\infty = \max_i |x_i|$$

$$\|x_1\|_\infty = 10$$

$$\|x_2\|_\infty = 5$$

b)



$\|x_1\|_1$ is Manhattan distance, shortest distance without moving diagonally.

$\|x_2\|_2$ is shortest distance from point to point:

$\|x_1\|_\infty$ is largest vector component: Pythagoras max(width, length) (absolute value)

b) Same for x_2. Not gonna attempt 3D drawing.

c) $y \in \mathbb{R}^n$, $\|y\|_1 \geq \|y\|_2$ proof

$$\|y\|_1 \geq \|y\|_2$$

$$\sum_{i=1}^{n} |y_i| \geq \sqrt{\sum_{i=1}^{n} |y_i|^2}$$

$$\left(\sum_{i=1}^{n} |y_i|\right)^2 \geq \sum_{i=1}^{n} |y_i|^2$$

Expand left side

$$\sum_{i=1}^{n} |y_i|^2 + \text{non-negative terms} \geq \sum_{i=1}^{n} |y_i|^2$$

Therefore $\|y\|_1 \geq \|y\|_2$ Q.E.D

d) $$A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 0 \\ 1 & 2 & 0 \end{bmatrix}$$

$$\|A\|_1 = \max_j \sum_{i=1}^{m} |a_{ij}| = |1| + |2| + |2| = 5$$

$$\|A\|_2 = \sqrt{\lambda_{max}(A^T A)}$$

$$A^T A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 2 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 0 \\ 1 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 6 & 6 & -2 \\ 6 & 9 & -1 \\ -2 & -1 & 1 \end{bmatrix}$$

$$\det(A^T A - \lambda I) = 0$$

$$\det\left(\begin{bmatrix} 6-\lambda & 6 & -2 \\ 6 & 9-\lambda & -1 \\ -2 & -1 & 1-\lambda \end{bmatrix}\right) = 0$$

$-\lambda(\lambda^2 - 16\lambda + 28) = 0 \Rightarrow -\lambda(\lambda-2)(\lambda-14) = 0$

$\lambda_1 = 0, \quad \lambda_2 = 2, \quad \lambda_3 = 14$

$\|A\|_2 = \sqrt{14}$

d) $\|A\|_\infty = \max_i \sum_{j=1}^{n} |a_{ij}| = |2| + |1| + |-1| = 4$

e) $\|Ax\|_p \leq \|A\|_p \|x\|_p$

$c_1 \|x\|_{q_1} \leq \|x\|_{q_2} \leq c_2 \|x\|_{q_1}$, where $c_1, c_2 > 0$

Show they're true for A and $x = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$ using $p=2$, $q_1=1$, $q_2=\infty$. Specify c_1, c_2.

1. $\|Ax\|_p \leq \|A\|_p \|x\|_p$

$\|Ax\|_2 \leq \|A\|_2 \|x\|_2$

$$\|Ax\|_2 = \left\| \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 0 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} \right\|_2 = 3\sqrt{3}$$

$3\sqrt{3} \leq \sqrt{14} \cdot \sqrt{14}$

$3\sqrt{3} \leq 14$

2. $c_1 \|x\|_1 \leq \|x\|_\infty \leq c_2 \|x\|_1$

$c_1 \cdot 6 \leq 3 \leq c_2 \cdot 6$

$c_1 = \frac{1}{2}$, $c_2 = 1$

f) $f_1(t) = \sqrt{t}$

$f_2(t) = \frac{1}{(t+1)^2}$

Decide if $f_i : [0, \infty) \to \mathbb{R}$ is L_∞, L_2, L_1

$f \in L_p \iff \|f\|_{L_p}$ is bounded. $\|f\|_{L_p} \leq c$

1. $\|f_1\|_{L_\infty} = \sup_{a \leq t \leq b} |f(t)| = \sup_{t \geq 0} |\sqrt{t}| = \infty$ Not in L_∞

$$\|f_1\|_{L_2} = \left(\int_a^b |f(\tau)|^p d\tau \right)^{\frac{1}{p}} = \left(\int_0^\infty |\sqrt{\tau}|^2 d\tau \right)^{\frac{1}{2}}$$

$= \infty$ Not in L_2

$$\|f_1\|_{L_2} = \left(\int_0^\infty |\sqrt{\tau}| \, d\tau \right) = \left[\frac{2}{3} \tau^{\frac{3}{2}} \right]_0^\infty = (\infty) - 0$$

f_1 not in L_1

2. $f_2(t) = \frac{1}{(t+1)^2}$

$$\|f_2\|_{L_\infty} = \sup_{t \geq 0} \left| \frac{1}{(t+1)^2} \right| = 1 \implies f_2 \in L_\infty [0, \infty)$$

$$\|f_2\|_{L_2} = \left(\int_0^\infty \left| \frac{1}{(t+1)^2} \right|^2 dt \right)^{\frac{1}{2}} = \sqrt{\left[-\frac{1}{3} (t+1)^{-3} \right]_0^\infty}$$

$= \sqrt{\frac{1}{3}} \implies f_2 \in L_2 [0, \infty)$

$$\|f_2\|_{L_1} = \int_0^\infty \left| \frac{1}{(t+1)^2} \right| dt = \left[-\frac{1}{t+1} \right]_0^\infty = -1 + 1 = 1$$

$f_2 \in L_1 [0, \infty)$

Problem 2

a) $f(x) = x^{\frac{1}{3}}$

Must exist constant $L \geq 0$ such that $\|f(x) - f(y)\| \leq L\|x - y\|$ for all x, y

$$\frac{\|f(x) - f(y)\|}{\|x - y\|} \leq L$$

We fix $y = 0$:

$$\frac{\|f(x) - f(0)\|}{\|x - 0\|} = \frac{|x^{\frac{1}{3}}|}{|x|} = \frac{1}{|x|^{\frac{2}{3}}} \leq L$$

When $x \to 0$, $\frac{1}{|x|^{\frac{2}{3}}} \to \infty \Rightarrow$ no L exists

Therefore it isnt globally Lipschitz

f is locally Lipschitz on $(-\infty, 0) \cup (0, \infty)$ because $f(x)$ is continuous on the interval and $\frac{\partial f}{\partial x}$ exists and is continuous on $(-\infty, 0) \cup (0, \infty)$ (Lemma 3.2)

b) $f(x) = \begin{bmatrix} x_2 \\ -\frac{g}{L}\sin(x_1) - \frac{k}{m}x_2 \end{bmatrix}$

Locally Lipschitz on state space? Check Lemma 3.2

1. f is continuous on $\mathbb{R}^2$

2. $\frac{\partial f}{\partial x}$ exists and is continuous for $x \in \mathbb{R}^2$

$$\frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{L}\cos x_1 & -\frac{k}{m} \end{bmatrix}$$

f is locally Lipschitz on $\mathbb{R}^2$

Globally Lipschitz:

Lemma 3.1

$\left\| \frac{\partial f}{\partial x} \right\| \leq L \Rightarrow$ Lipschitz on $\mathbb{R}^2$

$$\left\| \frac{\partial f}{\partial x} \right\|_1 = \max\left(\frac{g}{L}|\cos x_1|, 1 + \frac{k}{m} \right) \leq L$$

$|\cos x_1| \leq 1 \Rightarrow L = \max\left(\frac{g}{L}, 1 + \frac{k}{m} \right)$

f is globally Lipschitz because we can choose this L

By theorem 3.1 and 3.2 we have
local existence, global existence and uniqueness.

b) $f(x) = \begin{bmatrix} x_2 \\ -x_1 \frac{k}{m} - \frac{c}{m} x_2 + \frac{1}{m}\eta(x_1, x_2) \end{bmatrix}$

$$\eta(x_1, x_2) = \begin{cases} \mu_k mg \, sign(x_2) & \text{for } |x_2| > 0 \\ -k x_1 & \text{for } x_2 = 0 \text{ and } |x_1| \leq \frac{\mu_s mg}{k} \\ -\mu_s mg \, sign(x_1) & \text{for } x_2 = 0 \text{ and } |x_1| > \frac{\mu_s mg}{k} \end{cases}$$

Lemma 3.2
- If f is continuous and $\frac{\partial f}{\partial x}$ exists and is continuous, then f is locally Lipschitz

$\eta(0,0) = 0$

If $x_2 > 0$: $\eta(0, x_2) = \mu_k mg$

$\lim_{x_2 \to 0^+} \eta(0, x_2) = \mu_k mg \neq 0$

η is not continuous in $(0,0)$ $\Rightarrow$ f is discontinuous

Therefore it is not locally Lipschitz on whole s.s
Because it's not locally Lipschitz, then it's not globally Lipschitz either.

We cannot conclude local/global existence and uniqueness using Theorem 3.1 because f doesn't satisfy Lipschitz condition

c) $f(x) = \begin{bmatrix} \\ -x_1 + \varepsilon(1-x_1^2)x_2 \end{bmatrix}$

$$\frac{\partial f}{\partial x} = \begin{bmatrix} 0 & 1 \\ -2\varepsilon x_2 x_1 - 1 & \varepsilon(1-x_1^2) \end{bmatrix}$$

Lemma 3.2: f is continuous and $\frac{\partial f}{\partial x}$ exists and is continuous, therefore f is locally Lipschitz on $\mathbb{R}^2$

Lemma 3.1 $\left\| \frac{\partial f}{\partial x} \right\| \leq L \Rightarrow$ Globally Lipschitz

$$\left\| \frac{\partial f}{\partial x} \right\|_\infty = \max\left(|0| + |1|, \left| -1 - 2\varepsilon x_2 x_1 \right| + \left| \varepsilon(1-x_1^2) \right| \right)$$

If $x_2 = 0$ and $x_1 \to \infty$

$$\left\| \frac{\partial f}{\partial x} \right\|_\infty = \max\left(1, 1 + \left| \varepsilon(1-x_1^2) \right| \right) \to \infty$$

There is no L such that $\left\| \frac{\partial f}{\partial x} \right\| \leq L$
Therefore f is not globally Lipschitz

Locally Lipschitz $\Rightarrow$ exists local unique solution for every x_0

We cannot conclude anything about global existence and uniqueness

c) Show $\dot{x} = Ax$, $x \in \mathbb{R}^n$ is globally Lipschitz

Lipschitz condition

$$\|f(x) - f(y)\| = \|Ax - Ay\| = \|A(x-y)\|$$

$$\|A(x-y)\| \leq \|A\| \cdot \|x-y\|$$

$$\|f(x) - f(y)\| \leq \|A\| \cdot \|x-y\|$$

$L = \|A\|_p$ would work

Therefore it's globally Lipschitz

Choice of norm only changes L, not the global Lipschitz property.

Problem 3

a) $\dot{x} = x(\alpha - \beta y)$ $\quad x>0, y=0$

$\dot{y} = -y(\gamma - \delta x)$

$\dot{x} = \alpha x \Rightarrow x = x_0 e^{\alpha t}$ Rabbit population explodes

$\dot{y} = 0 \quad\quad y = 0$

b) $\dot{x} = 0 \quad\quad x = 0$

$\dot{y} = -\gamma y \Rightarrow y = y_0 e^{-\gamma}$ Foxes die

c) $x(\alpha - \beta y) = 0$

$-y(\gamma - \delta x) = 0$

$y = 0 \quad x = 0$

$y = \frac{\alpha}{\beta} \quad x = \frac{\gamma}{\delta}$

c) We're going to
- Linearize about equilibrium
- Find eigenvalues $\lambda(A)$
- Classify the eq. points

$$A(x,y) = \begin{bmatrix} \alpha - \beta y & -\beta x \\ \delta y & -\gamma + \delta x \end{bmatrix}$$

$$A(0,0) = \begin{bmatrix} \alpha & 0 \\ 0 & -\gamma \end{bmatrix} \quad \lambda_1 = \alpha > 0, \quad \lambda_2 = -\gamma < 0$$

Saddle point: $\lambda_2 < 0 < \lambda_1$

$$A\left(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}\right) = \begin{bmatrix} 0 & -\frac{\beta\gamma}{\delta} \\ \frac{\delta\alpha}{\beta} & 0 \end{bmatrix}$$

$$\det(\lambda I - A) = 0$$

$$\lambda^2 + \alpha\gamma = 0 \Rightarrow \lambda_{1,2} = \pm i\sqrt{\alpha\gamma}$$

$$Re(\lambda_i) = 0 \Rightarrow \text{Center}$$

d) $\dot{x} = Ax$

$Ax^* = 0$

Two cases for null space

I) $\ker(A) = \{0\}, \quad x^* = 0$

II) $\ker(A) \neq \{0\}$

$A(cx^*) = cAx^* = 0 \Rightarrow$ infinitely many equilibrium points that are non-isolated. They're scaled versions of each other

Multiple isolated points not possible

e) The origin is a saddle point, meaning the trajectories tend to move away unless $x = 0$ and foxes die.
As mentioned earlier when $y = 0$ and $x > 0$ the rabbit population explodes.

The second equilibrium is in $\left(\frac{457}{6}, \frac{63}{5}\right)$ which is a center. Other trajectories revolve around this point.

f) No, because if you try to cross axes, you will either hit $x = 0$ or $y = 0$

$x = 0 \Rightarrow \dot{x} = 0$

$y = 0 \Rightarrow \dot{y} = 0$

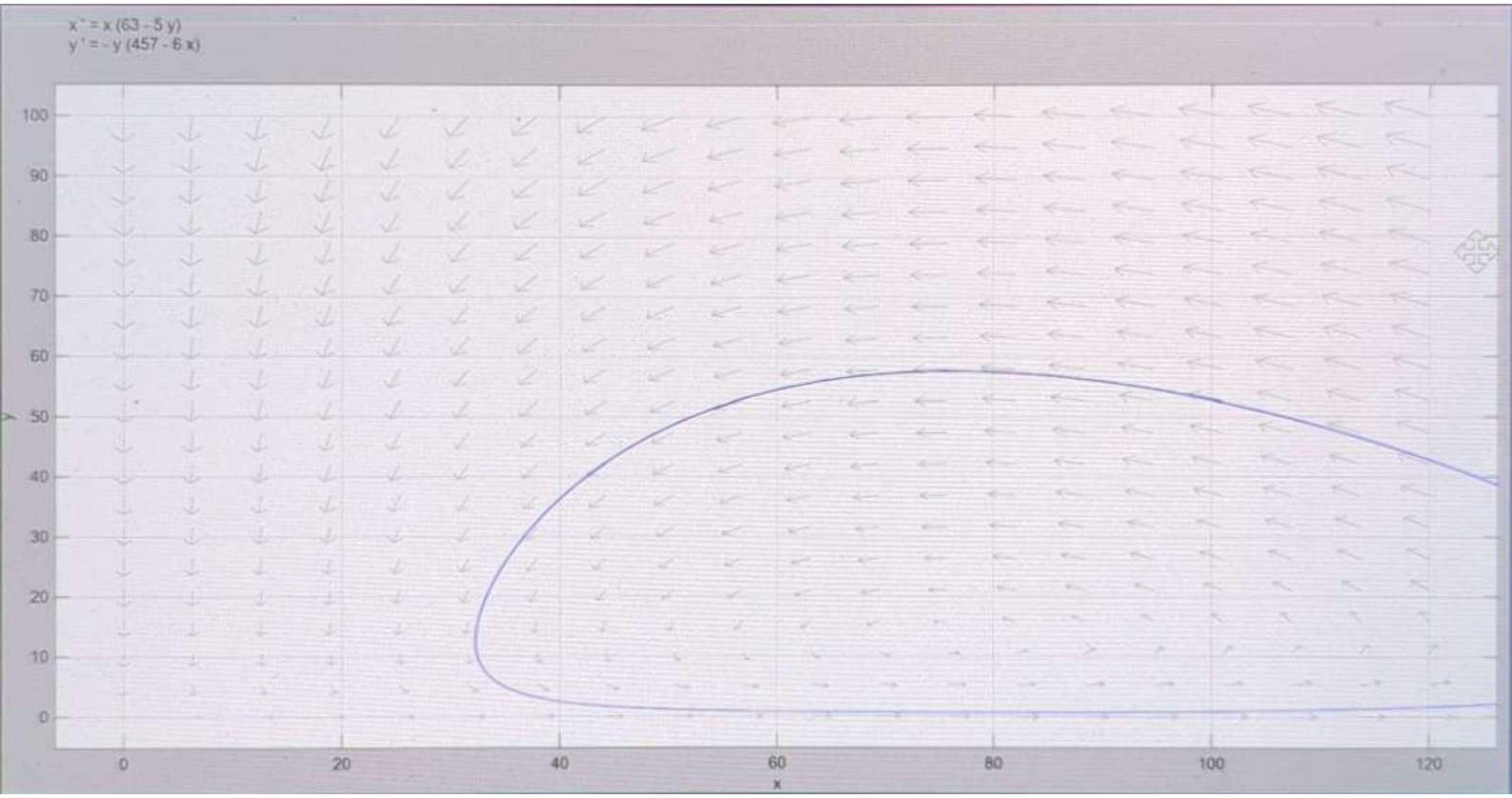
They will stop changing and not go below 0

g) $\nabla \cdot f = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} = \alpha - \beta y - \gamma + \delta x$

$D_1 = \{(x, y) : a - \beta y - \gamma + \delta x > 0\}$

or

$D_2 = \{(x, y) : a - \beta y - \gamma + \delta x < 0\}$


x' = x (63 - 5 y)
y' = - y (457 - 6 x)
100
90
80
70
60
50
40
30
20
10
0
y
0
20
40
60
80
100
120
x

Problem 4

a) $\dot{x}_1 = (2x_2 - 1)x_1$

$\dot{x}_2 = \cos(x_1) - (x_2 + x_2^2 + x_2^4)x_2$

$\nabla \cdot f = 2x_2 - 1 - 5x_2^4 + 3x_2^2 - 2x_2 = -5x_2^4 - 3x_2^2 - 1$

$\nabla \cdot f < 0 \Rightarrow$ No periodic orbits on $\mathbb{R}^2$

b) Index theorem because Bendixson is inconclusive

C is a periodic orbit $\Rightarrow \sum_i I(\text{eq point } i)$ within $C = +1$

I(node, focus, center) $= +1$

I(saddle point) $= -1$

We use $\sum_i I \neq 1 \Rightarrow$ C is not periodic orbit

Find equilibrium points:

$\dot{x}_1 = x_2 + x_1^2 x_2 = x_2(x_1^2 + 1) = 0$

$\dot{x}_2 = x_1 + x_1^3 = x_1(x_1^2 + 1) = 0$

$\Rightarrow x^* = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\frac{\partial f}{\partial x} = \begin{bmatrix} 2x_1 x_2 & 1 + x_1^2 \\ 1 + 3x_1^2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\det\left(\lambda I - \frac{\partial f}{\partial x}\right) = \lambda^2 - 1 = 0 \Rightarrow \lambda_{1,2} = \pm 1$$

(0,0) is saddle point

Two cases:

1. C does not enclose (0,0): $\sum I = 0$
2. C does enclose (0,0): $\sum I = -1$

No case gives $\sum I = 1$ so no periodic orbit exists in $\mathbb{R}^2$

Problem 5

$\dot{x}_1 = x_2 + \alpha x_1(\beta^2 - x_1^2 - x_2^2)$ $\quad \alpha, \beta \geq 0$

$\dot{x}_2 = -x_1 + \alpha x_2(\beta^2 - x_1^2 - x_2^2)$

$x_1 = r\cos\theta \quad x_2 = r\sin\theta$

$\frac{x_2}{x_1} = \frac{r\sin\theta}{r\cos\theta} \Rightarrow \theta = \tan^{-1}\left(\frac{x_2}{x_1}\right)$

$x_1^2 + x_2^2 = r^2(\cos^2\theta + \sin^2\theta) \Rightarrow r = \pm\sqrt{x_1^2 + x_2^2}$

$z = \Psi(x)$

$\begin{bmatrix} r \\ \theta \end{bmatrix} = \begin{bmatrix} \sqrt{x_1^2 + x_2^2} \\ \tan^{-1}\left(\frac{x_2}{x_1}\right) \end{bmatrix}$

$\dot{r} = \frac{x_1\dot{x}_1 + x_2\dot{x}_2}{\sqrt{x_1^2 + x_2^2}} = \frac{x_1\dot{x}_1 + x_2\dot{x}_2}{r} = \frac{\alpha r^2(\beta^2 - r^2)}{r}$

$\dot{\theta} = \frac{x_1\dot{x}_2 - x_2\dot{x}_1}{x_1^2 + x_2^2} = \frac{-r^2}{r^2} = -1$

6. $\dot{x}_1 = -ax_1 - x_2^2$ $\quad a,b,c > 0$

$\dot{x}_2 = bx_1x_2 + cx_2$

a) $\frac{\partial f}{\partial x} = \begin{bmatrix} -a & -2x_2 \\ bx_2 & bx_1 + c \end{bmatrix}$

b) $-ax_1 - x_2^2 = 0$

$bx_1x_2 + cx_2 = 0$

$x_2(bx_1 + c) = 0$

$x^*_{eq1} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\quad x^*_{eq2} = \begin{bmatrix} -\frac{c}{b} \\ \sqrt{\frac{ac}{b}} \end{bmatrix}$ $\quad x^*_{eq3} = \begin{bmatrix} -\frac{c}{b} \\ -\sqrt{\frac{ac}{b}} \end{bmatrix}$

$\frac{\partial f}{\partial x}(0,0) = \begin{bmatrix} -a & 0 \\ 0 & c \end{bmatrix}$ $\quad \lambda_1 = -a < 0$, $\lambda_2 = c > 0 \Rightarrow$ Saddle point

$\frac{\partial f}{\partial x}(x^*) = \begin{bmatrix} -a & -2x_2^* \\ bx_2^* & 0 \end{bmatrix}$

$\det\left(\lambda I - \frac{\partial f}{\partial x}\right) = \lambda^2 + a\lambda + 2b(x_2^*)^2 = 0$

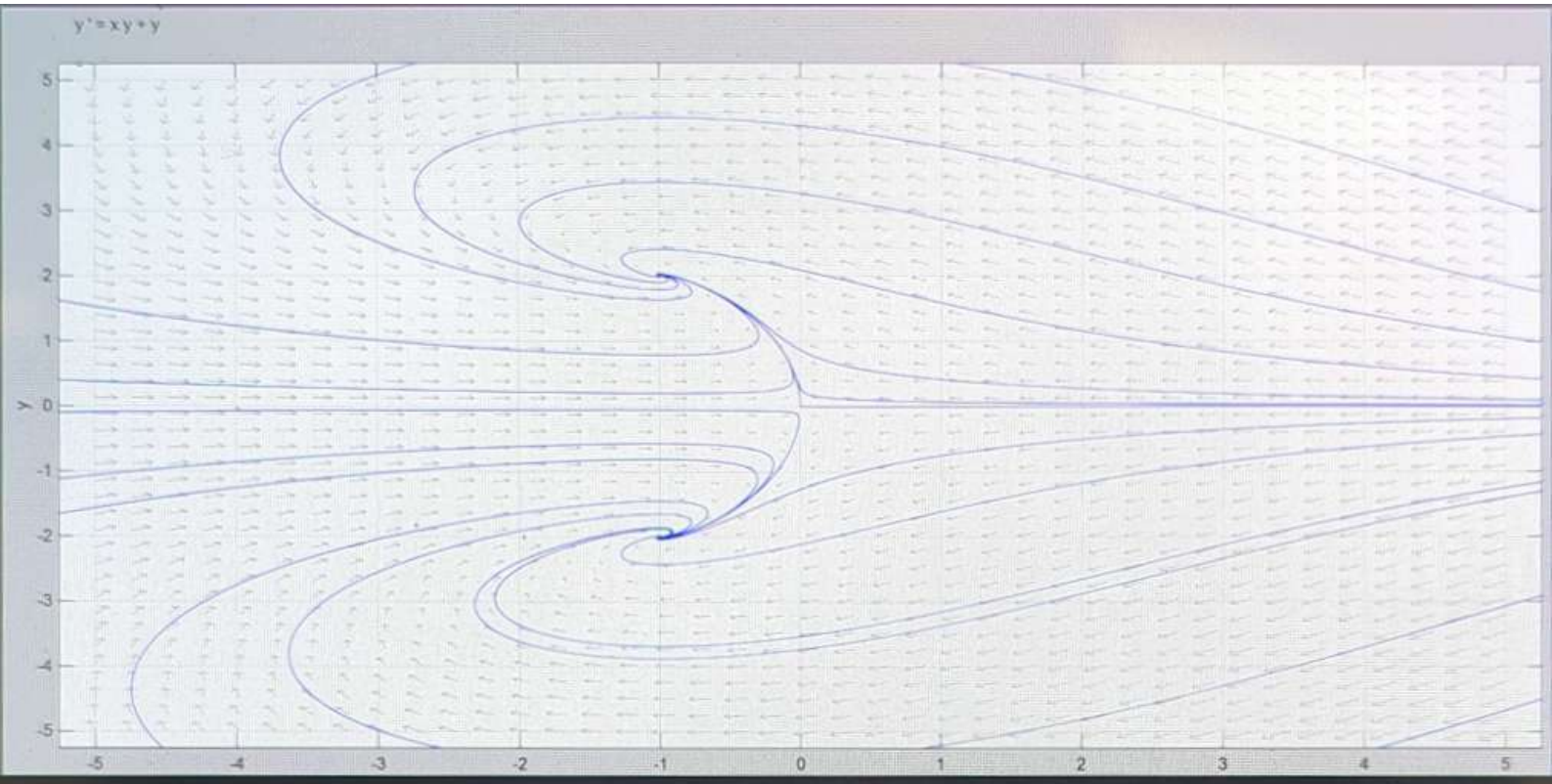
$\lambda_{1,2} = \frac{-a \pm \sqrt{a^2 - 8ac}}{2}$

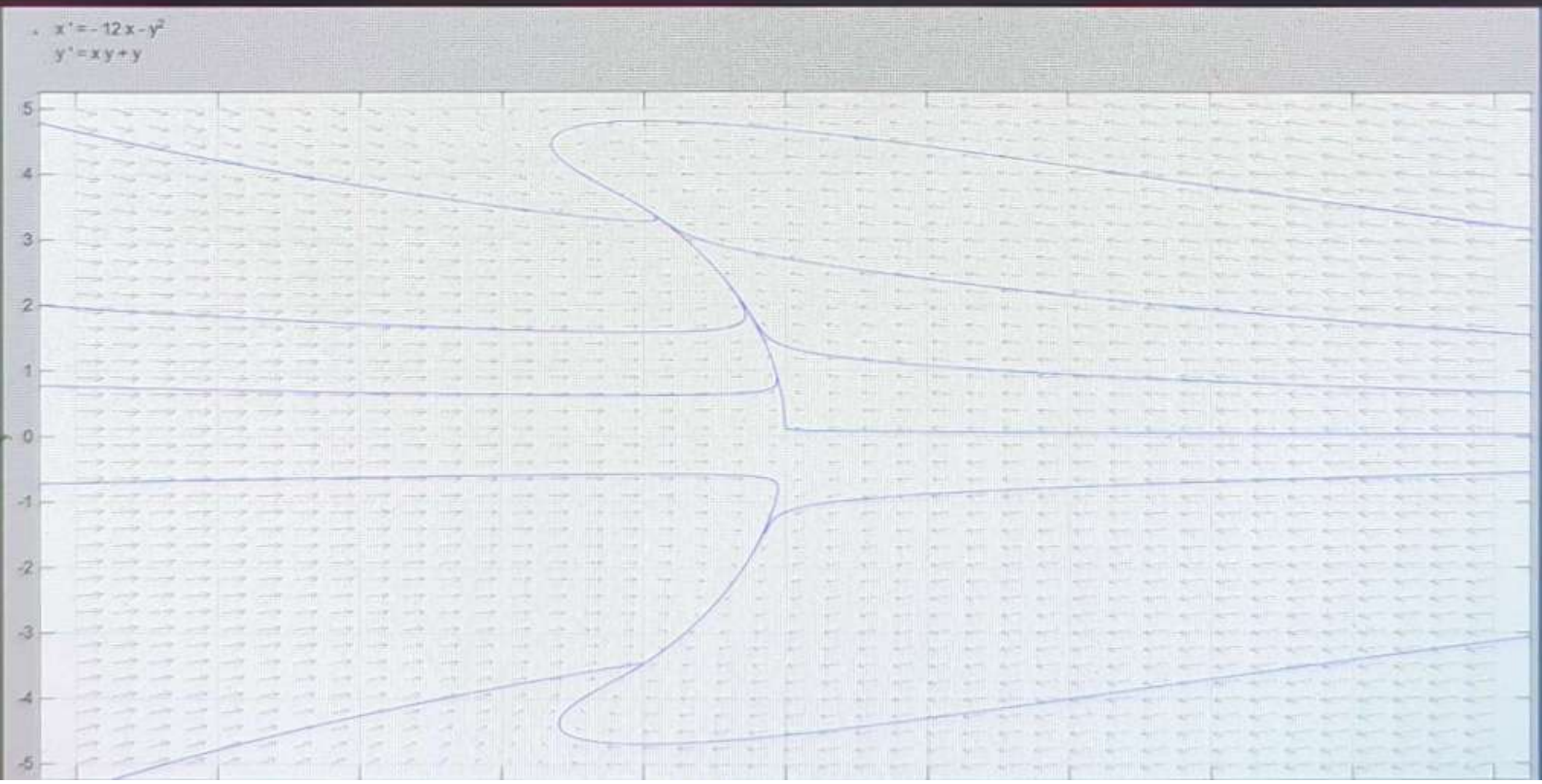
For x^*_{eq2} and x^*_{eq3}:

Stable node if $a > 8c$

Stable degenerate node if $a = 8c$

Stable focus if $a < 8c$


y' = xy + y
y
5
4
3
2
1
0
-1
-2
-3
-4
-5
-5
-4
-3
-2
-1
0
1
2
3
4
5


x' = -12x - y²
y' = xy + y
5
4
3
2
1
0
-1
-2
-3
-4
-5

c) a) We see that trajectories avoid $(0,0)$ because it's a saddle point, Except for those where $x_2(0) = 0$

The other trajectories converge to $(-1, 2)$ and $(-1, -2)$ because they're stable foci. $a = 4 < 8c = 8$.
They kinda spiral around the equilibrium points.

b) $(0,0)$ is still saddle
$(-1, \pm\sqrt{12})$ is stable node because $a = 12 > 8c = 8$

Problem 7

a) $M = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ $P_{11} = 1 \geq 0$ $P_{22} = 1 \geq 0$
$\det(M) = 0 \geq 0$

M is PSD

$x^T M x \geq 0$

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \geq 0$$

$x_1^2 - 2x_1x_2 + x_2^2 \geq 0$
$2x_1x_2 \leq x_1^2 + x_2^2$

b) $M^+ = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ $P_{11} = 1 \geq 0$ $P_{22} = 1 \geq 0$
$\det(M^+) = 0 \geq 0$
M^+ is PSD

$x^T M^+ x = x_1^2 + 2x_1x_2 + x_2^2 \geq 0$
$-2x_1x_2 \leq x_1^2 + x_2^2$
Part a: $2x_1x_2 \leq x_1^2 + x_2^2$
$\Rightarrow 2|x_1x_2| \leq x_1^2 + x_2^2$

c) $M_\varepsilon^- = \begin{bmatrix} \varepsilon & -1 \\ -1 & \frac{1}{\varepsilon} \end{bmatrix}$ $\varepsilon > 0$ $\frac{1}{\varepsilon} > 0$

$\det(M_\varepsilon^-) = 0 \geq 0$

M_ε^- is PSD

$x^T M_\varepsilon^- x \geq 0$

$\varepsilon x_1^2 - 2x_1 x_2 + \frac{1}{\varepsilon} x_2^2 \geq 0$

$2x_1 x_2 \leq \varepsilon x_1^2 + \frac{1}{\varepsilon} x_2^2$

Same for $M_\varepsilon^+ = \begin{bmatrix} \varepsilon & 1 \\ 1 & \frac{1}{\varepsilon} \end{bmatrix}$ $M \succeq 0$

$\rightsquigarrow -2x_1 x_2 \leq \varepsilon x_1^2 + \frac{1}{\varepsilon} x_2^2$

Together: $2|x_1 x_2| \leq \varepsilon x_1^2 + \frac{1}{\varepsilon} x_2^2 \quad \varepsilon > 0$

d) $x_1 x_2 \leq |x_1 x_2| \leq \frac{\varepsilon}{2} x_1^2 + \frac{1}{2\varepsilon} x_2^2$

$\varepsilon = 12$: $x_1 x_2 \leq 6x_1^2 + \frac{1}{24} x_2^2$

Therefore:

$x_1 x_2 - 8x_1 - \frac{1}{20} x_2^2 \leq -2x_1^2 - \frac{1}{120} x_2^2 < 0$

Problem 8

a)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1-(x_1^2+x_2^2) & 2 \\ -3 & 1-(x_1^2+x_2^2) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$g(x) = 1 - x_1^2 - x_2^2$

b) $\begin{bmatrix} g(x) & 2 \\ -3 & g(x) \end{bmatrix} x = 0$

$\det\left(\begin{bmatrix} g(x) & 2 \\ -3 & g(x) \end{bmatrix}\right) = g(x)^2 + 6 > 0$

Matrix is invertible which means $(0,0)$ is only equilibrium

Linearization matrix:

$A = \begin{bmatrix} 1 & 2 \\ -3 & 1 \end{bmatrix} \quad \lambda = 1 \pm i\sqrt{6}$

$(0,0)$ is unstable focus because $\mathrm{Re}(\lambda) > 0$

c) $h(x) = x_1^2 + x_2^2 - 2$

$\dot{h} = \frac{\partial h}{\partial x_1}\dot{x}_1 + \frac{\partial h}{\partial x_2}\dot{x}_2 = 2x_1\dot{x}_1 + 2x_2\dot{x}_2$

$\dot{x}_1 = g(x)x_1 + 2x_2, \quad \dot{x}_2 = -3x_1 + g(x)x_2$

with $g(x) = 1 - x_2^2 - x_2^2$

we get

$\dot{h} = 2x_1(g(x)x_1 + 2x_2) + 2x_2(-3x_1 + g(x)x_2)$

$\dot{h} = 2g(x)(x_1^2 + x_2^2) - 2x_1x_2$

$x_1^2 + x_2^2 = 2$ on C and $g(x) = -1$

Therefore $\dot{h} = -4 - 2x_1x_2$ on C

d) $\dot{h}(x) = -4 - 2x_1x_2$

$2|x_1x_2| \leq x_1^2 + x_2^2$

$x_1^2 + x_2^2 = 2 \Rightarrow 2|x_1x_2| \leq 2$

$|x_1x_2| \leq 1$

$-2x_1x_2 \leq 2$

$\dot{h} = -4 - 2x_1x_2 \leq 2 < 0$

Since $\dot{h} < 0$ on the boundary, trajectories reaching C are pushed into M

e) Poincaré-Bendixon criterion:

$M = \{x \in \mathbb{R}^2 \mid h(x) \leq 0\}$ closed and bounded

Every trajectory in M stays in M

Only equilibrium is in the origin

It's an unstable focus with $\mathrm{Re}(\lambda) > 0$

Therefore M contains a periodic orbit